



Docker files and Docker Compose

What is a Dockerfile?

A **Dockerfile** is a text file containing **step-by-step instructions** on how to build a Docker image.

It defines:

- What base image to use
- What dependencies to install
- What files to copy
- What ports to expose
- What command to run when the container starts

Think of a Dockerfile as **a recipe** for building your application image.



Why Do We Need a Dockerfile?

- Ensures **consistent environments** for all developers.
- Automates the creation of images.
- Makes builds **repeatable and portable**.
- Removes the “works on my machine” issue.



What is Docker Compose?

Docker Compose is a tool for running **multi-container applications** using a single YAML file.

Examples:

- App + Database
- Frontend + Backend + Redis
- Microservices setup

Instead of running multiple `docker run` commands manually, Compose manages everything with one file and one command.



Why Do We Need Docker Compose?

- To run **multiple containers together**.
- Simplifies network creation.
- Automates volume mounting, environment variables, dependencies.
- Easily share application definition (`docker-compose.yml`).

